

LECTURE 2

Linear regression

Foundations of Artificial Intelligence

SEMTM0051 · 2026/27

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Today's lecture

Regression and loss functions

Least-squares fitting

Gradient descent

Extensions

Today's lecture

Regression and loss functions

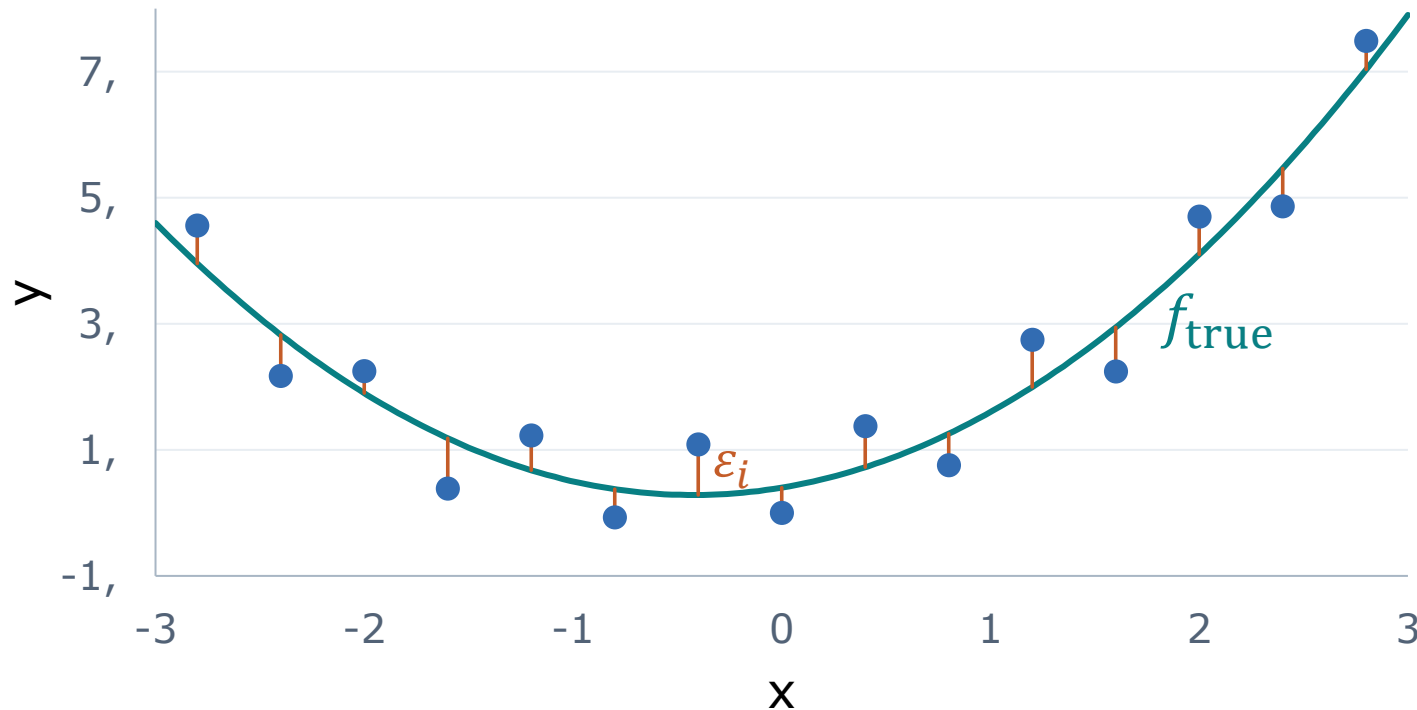
Least-squares fitting

Gradient descent

Extensions

Regression: predicting a numerical outcome

We observe N input–target pairs $(x_1, y_1), \dots, (x_N, y_N) \in \mathbb{R}^2$.



We model each outcome as a relationship plus noise.

$$y_i = f_{\text{true}}(x_i) + \varepsilon_i$$

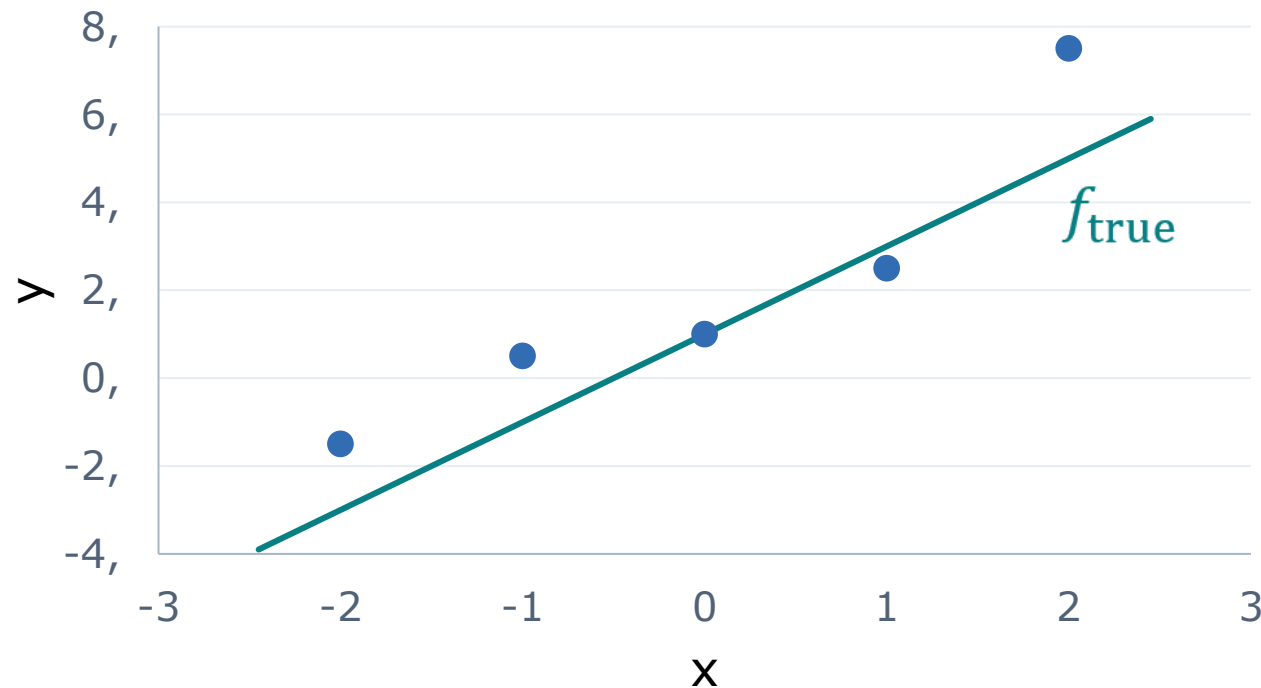
$$i = 1, \dots, N$$

The vertical segments represent the noise terms ε_i .

The relationship f_{true} and the noise terms ε_i are **unknown**.

An affine relationship

As a first special case, suppose the unknown relationship is **affine**.



$$f_{\text{true}}(x) = ax + b$$

$$y_i = ax_i + b + \varepsilon_i$$

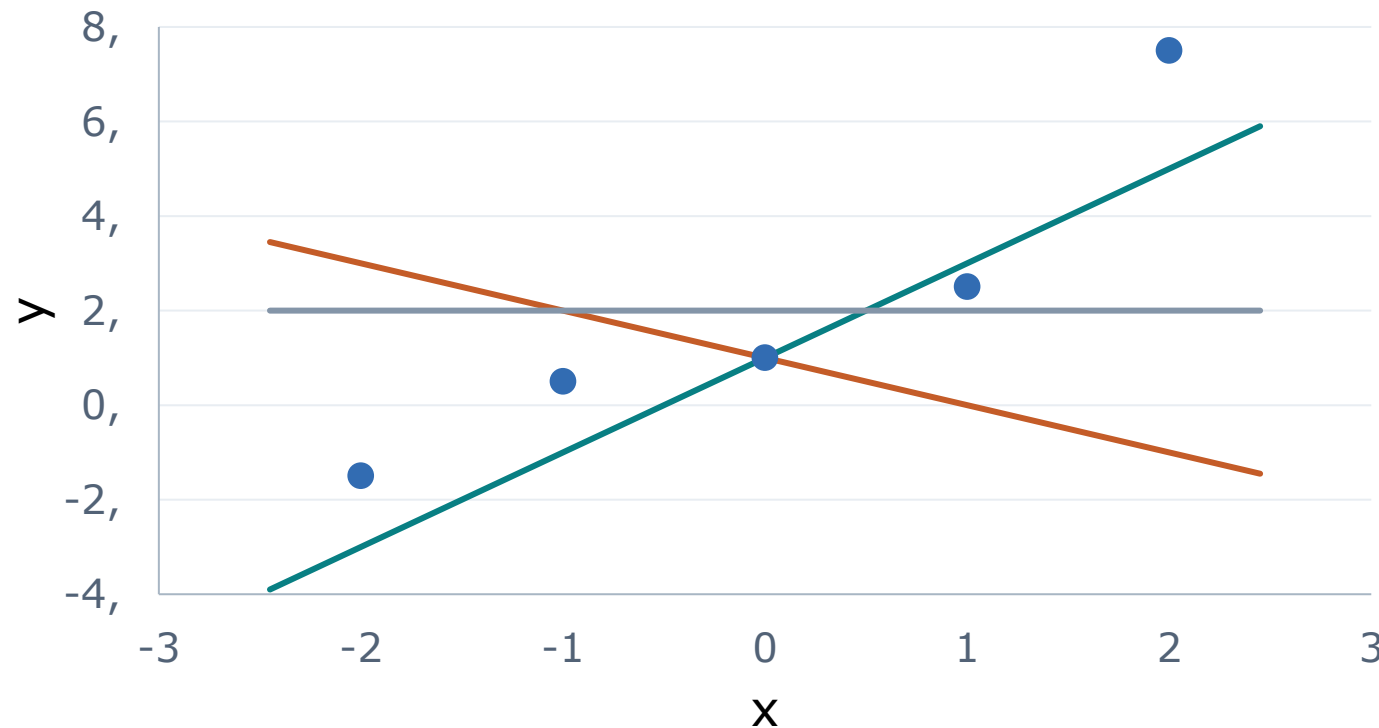
a : slope

b : intercept (also called **bias**)

The observed points **need not** lie exactly on the unknown line because of the ε_i

Several possible prediction lines

We do not know the true slope or intercept of f_{true} .
In general, there won't be any straight line that passes through all points.
So... which line should we use as our estimate?



$$f(x) = 1 + 2x$$

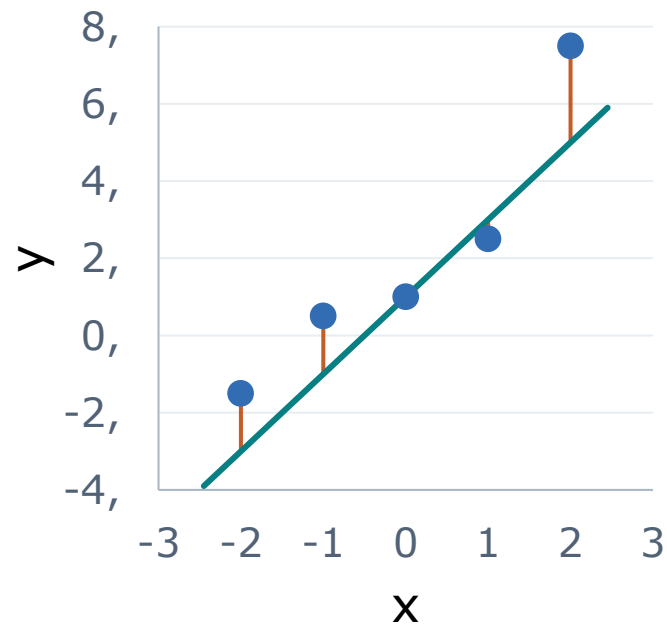
$$g(x) = 1 - x$$

$$h(x) = 2$$

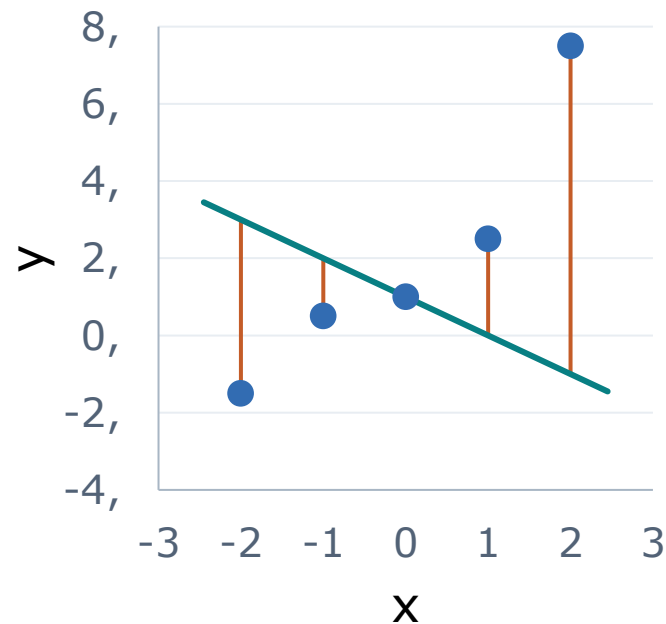
Comparing prediction errors

The vertical segments join each observation to its prediction.

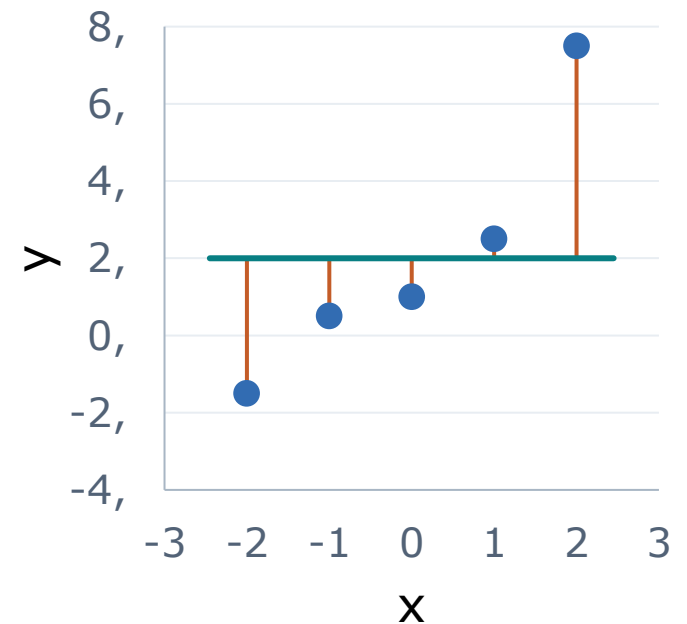
$$f(x) = 1 + 2x$$



$$g(x) = 1 - x$$



$$h(x) = 2$$



We look at the residuals: **residual = prediction – target**

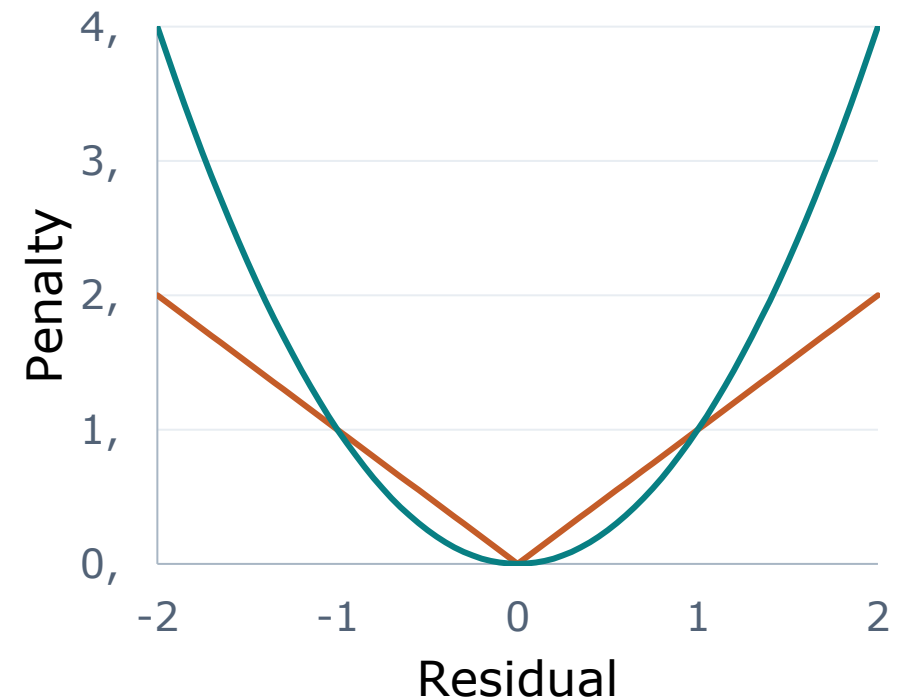
Loss functions

A **loss** assigns a **penalty** to one prediction $\hat{y}$ and its target y .

$$\ell_{\text{abs}}(\hat{y}, y) = |\hat{y} - y|$$

$$\ell_{\text{sq}}(\hat{y}, y) = (\hat{y} - y)^2$$

Squared loss is **differentiable** everywhere:
we can use differential calculus directly.
Absolute loss has a **kink** at zero.



Doubling the error doubles its absolute penalty and quadruples its squared penalty.

Empirical loss

Let θ collect the parameters of the predictor f_θ .

We add the losses over all observed pairs $(x_1, y_1), \dots, (x_N, y_N)$.

$$\mathcal{L}_{\text{abs}}(\theta) = \sum_{i=1}^N \ell_{\text{abs}}(f_\theta(x_i), y_i) = \sum_{i=1}^N |f_\theta(x_i) - y_i|$$

$$\mathcal{L}_{\text{sq}}(\theta) = \sum_{i=1}^N \ell_{\text{sq}}(f_\theta(x_i), y_i) = \sum_{i=1}^N (f_\theta(x_i) - y_i)^2$$

The data stay fixed. The parameters vary.

Choosing the loss

The loss should reflect the goal of the application.

Squared loss is **smooth** and convenient for differentiation.
It also gives large errors more influence.

Absolute loss is a valid alternative.
Its **corner at zero** needs a different treatment in some optimisation arguments.

We focus on squared loss in this lecture.

Comparing the candidates numerically

$$f(x) = 1 + 2x, \quad g(x) = 1 - x, \quad h(x) = 2$$

Observation	1	2	3	4	5	$\mathcal{L}_{\text{sq}}$
Input	-2	-1	0	1	2	
Target	-1.5	0.5	1	2.5	7.5	
Residual (f)	-1.5	-1.5	0	0.5	-2.5	11
Residual (g)	4.5	1.5	0	-2.5	-8.5	101
Residual (h)	3.5	1.5	1	-0.5	-5.5	46

The squared-error criterion prefers the first candidate.

This comparison does not yet identify the best line among all possible lines.

Today's lecture

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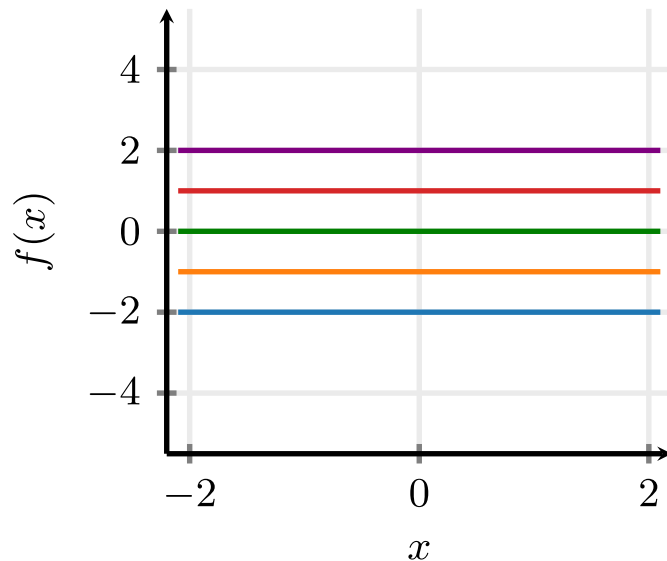
Finding a minimum

Fermat's theorem: minima are **stationary** points.
A differentiable objective has zero gradient at a local minimum.

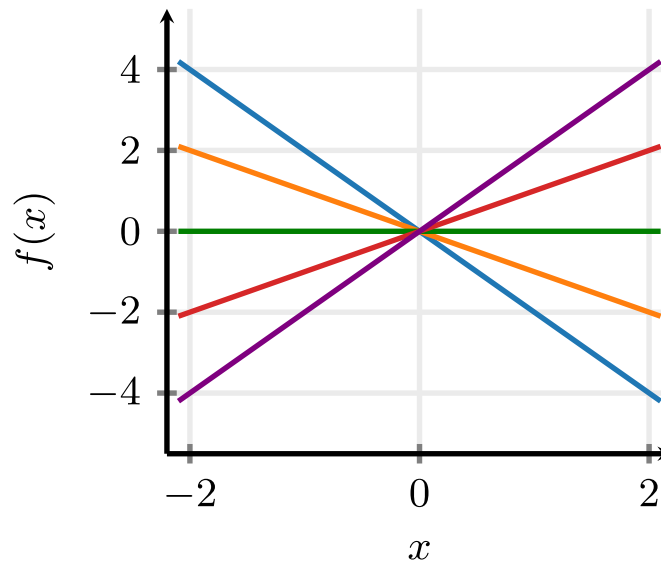
$$\nabla \mathcal{L}_{\text{sq}}(\boldsymbol{\theta}) = \mathbf{0}$$

We first find the stationary points, then check whether they are minima.

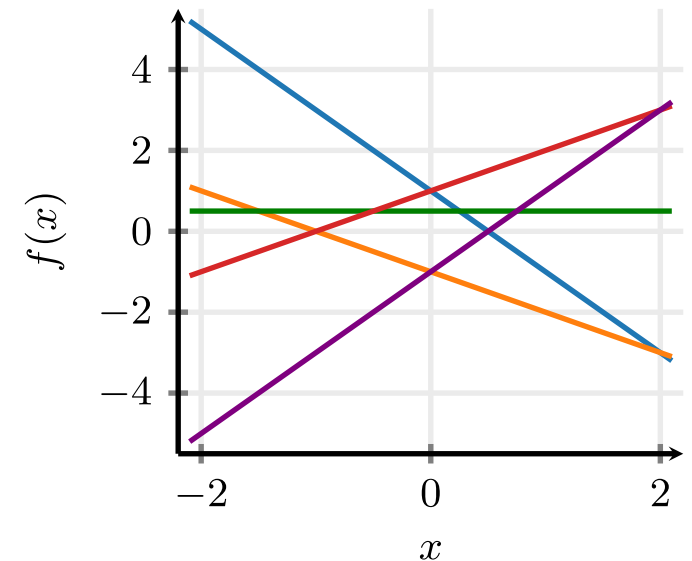
Constant: $f(x) = b$



Slope: $f(x) = ax$



Affine: $f(x) = b + ax$

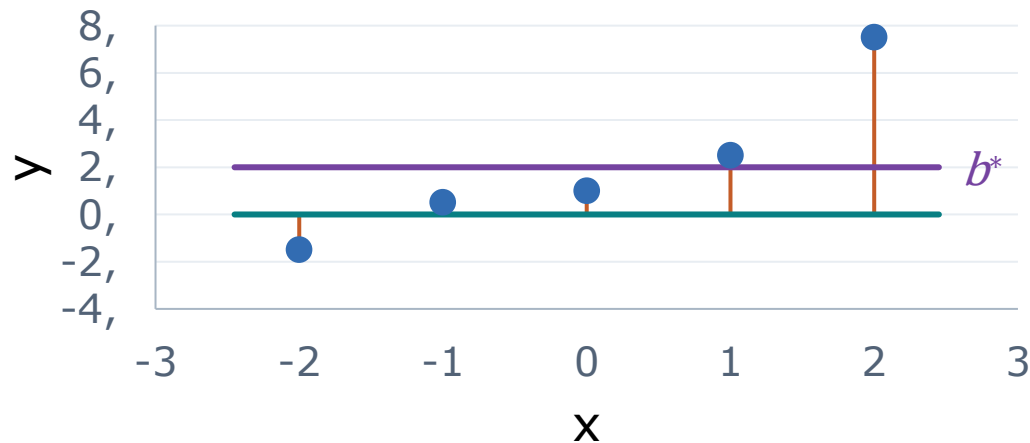


A constant predictor

Set $a = 0$. The predictor returns the same value b at every input.

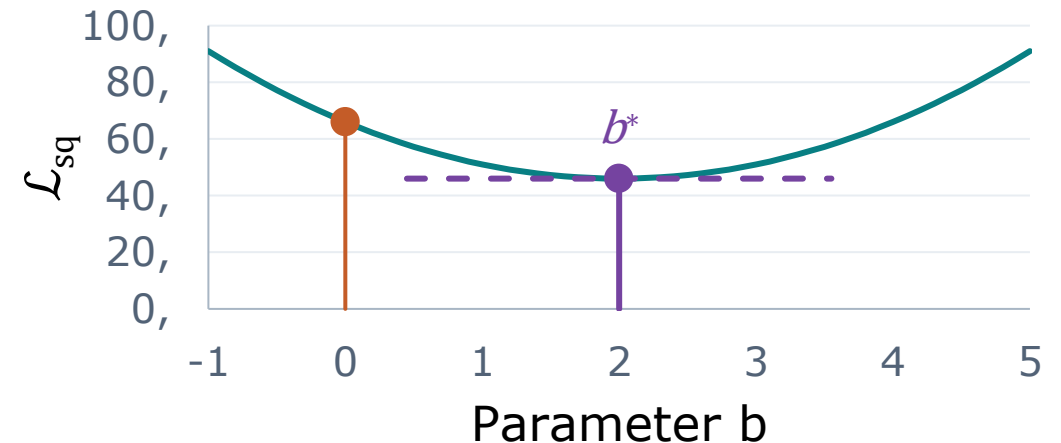
$$f_b^{\text{const}}(x) = b, \quad \mathcal{L}_{\text{sq}}^{\text{const}}(b) = \sum_{i=1}^N \ell_{\text{sq}}(f_b^{\text{const}}(x_i), y_i) = \sum_{i=1}^N (b - y_i)^2$$

Prediction as a function of input



How do we find b^* ?

Loss as a function of the parameter



b^* is a stationary point of $\mathcal{L}_{\text{sq}}^{\text{const}}$!

Differentiating the constant-predictor loss

$$\mathcal{L}_{\text{sq}}^{\text{const}}(b) = \sum_{i=1}^N (b - y_i)^2 = \sum_{i=1}^N (r_i(b))^2$$

where $r_i(b) = b - y_i$ is the i -th residual, so $r'_i(b) = 1$.

The **chain rule**, with outer function $u(r) = r^2$, and inner functions r_i gives

$$\begin{aligned} (\mathcal{L}_{\text{sq}}^{\text{const}})'(b) &= \frac{d}{db} \sum_{i=1}^N (r_i(b))^2 = \sum_{i=1}^N \frac{d}{db} (r_i(b))^2 = \sum_{i=1}^N u'(r_i(b)) r'_i(b) \\ &= \sum_{i=1}^N 2r_i(b) r'_i(b) = 2 \sum_{i=1}^N (b - y_i) = 2 \left(Nb - \sum_{i=1}^N y_i \right) \end{aligned}$$

The targets y_i are fixed. The parameter b does not depend on i .

The best constant is the sample mean

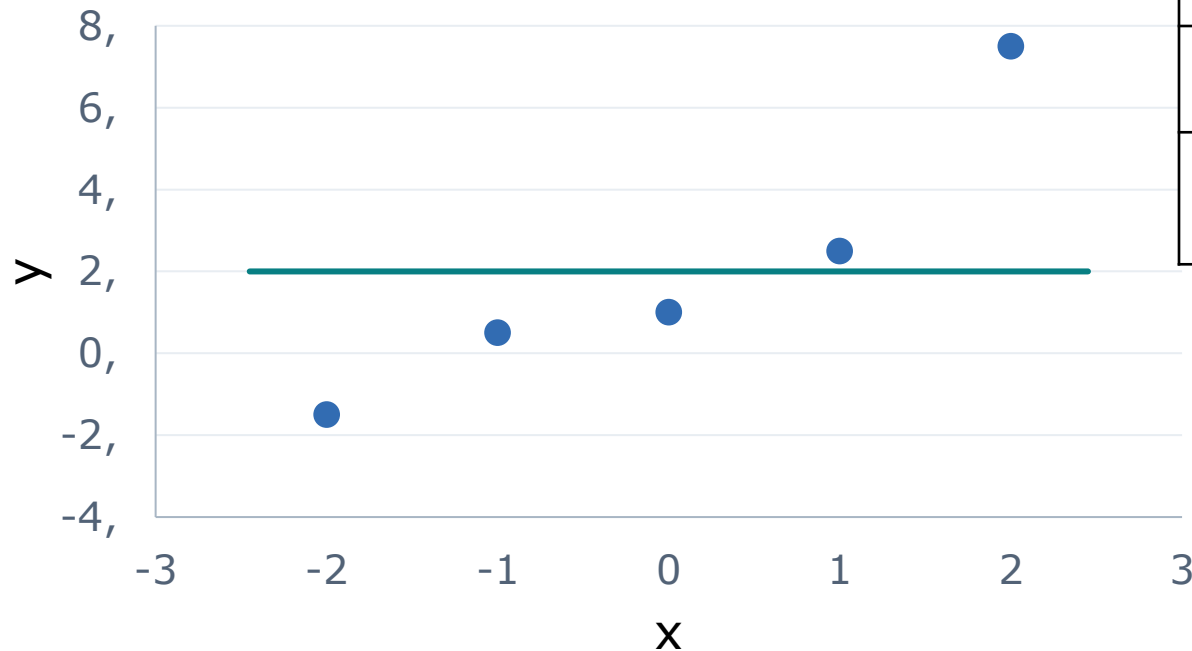
$$(\mathcal{L}_{\text{sq}}^{\text{const}})'(b) = 0 \quad \Leftrightarrow \quad Nb = \sum_{i=1}^N y_i$$

$$b^* = \frac{1}{N} \sum_{i=1}^N y_i = \bar{y}$$

The loss is a quadratic in b with positive leading coefficient N . Its graph is an upward-opening parabola, so a global minimum exists.

By Fermat, every minimiser must be stationary. There is only one stationary point, so it is the unique global minimiser.

The constant fit on our data



Obs	1	2	3	4	5	$\mathcal{L}_{\text{sq}}^{\text{const}}$
Input	-2	-1	0	1	2	
Target	-1.5	0.5	1	2.5	7.5	
Squared residual	12.25	2.25	1	0.25	30.25	46

$$b^* = \frac{-1.5 + 0.5 + 1 + 2.5 + 7.5}{5} = 2$$

$$\mathcal{L}_{\text{sq}}^{\text{const}}(b^*) = 46$$

**Exercise: keep the constant family and replace squared loss with
absolute loss**

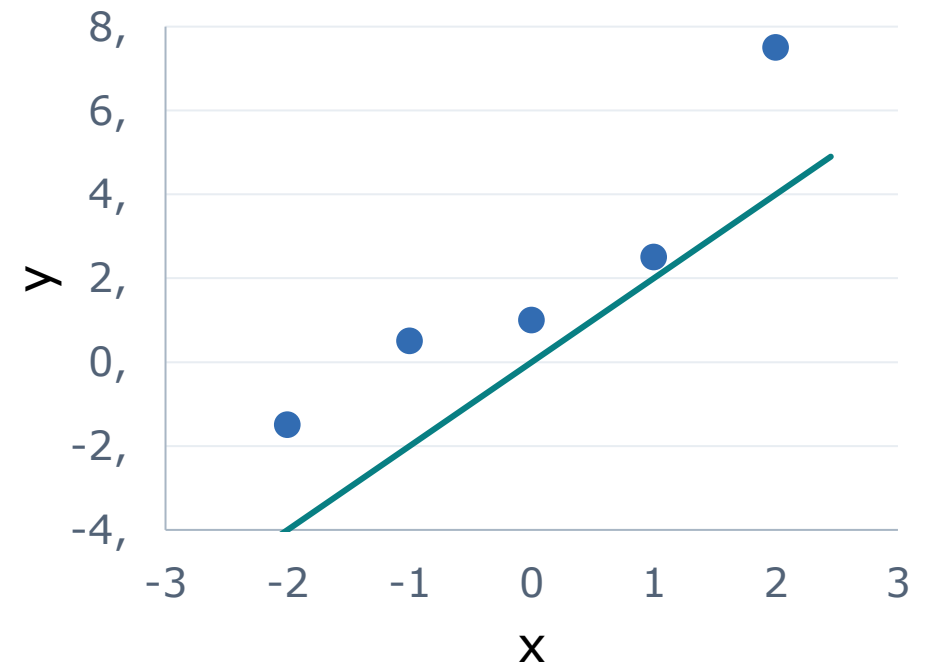
Which constant minimises the empirical loss?

A line through the origin

Set $b = 0$. The only adjustable parameter is the slope a .

$$f_a^{\text{slope}}(x) = ax$$

$$\begin{aligned}\mathcal{L}_{\text{sq}}^{\text{slope}}(a) &= \sum_{i=1}^N \ell_{\text{sq}}(f_a^{\text{slope}}(x_i), y_i) \\ &= \sum_{i=1}^N (ax_i - y_i)^2\end{aligned}$$



Every predictor in this family passes through the origin.

Differentiating with respect to the slope

$$\mathcal{L}_{\text{sq}}^{\text{slope}}(a) = \sum_{i=1}^N (ax_i - y_i)^2 = \sum_{i=1}^N r_i(a)^2$$

where $r_i(a) = ax_i - y_i$ is the i -th residual, so $r_i'(a) = x_i$.

The **chain rule**, with outer function $u(r) = r^2$, and inner functions r_i gives

$$\begin{aligned} \left(\mathcal{L}_{\text{sq}}^{\text{slope}}\right)'(a) &= \frac{d}{da} \sum_{i=1}^N (r_i(a))^2 = \sum_{i=1}^N \frac{d}{da} (r_i(a))^2 \\ &= \sum_{i=1}^N 2r_i(a)r_i'(a) = 2 \sum_{i=1}^N (ax_i - y_i)x_i = 2 \left(a \sum_{i=1}^N x_i^2 - \sum_{i=1}^N x_i y_i \right) \end{aligned}$$

Both x_i and y_i are fixed when we differentiate with respect to a .

The best slope

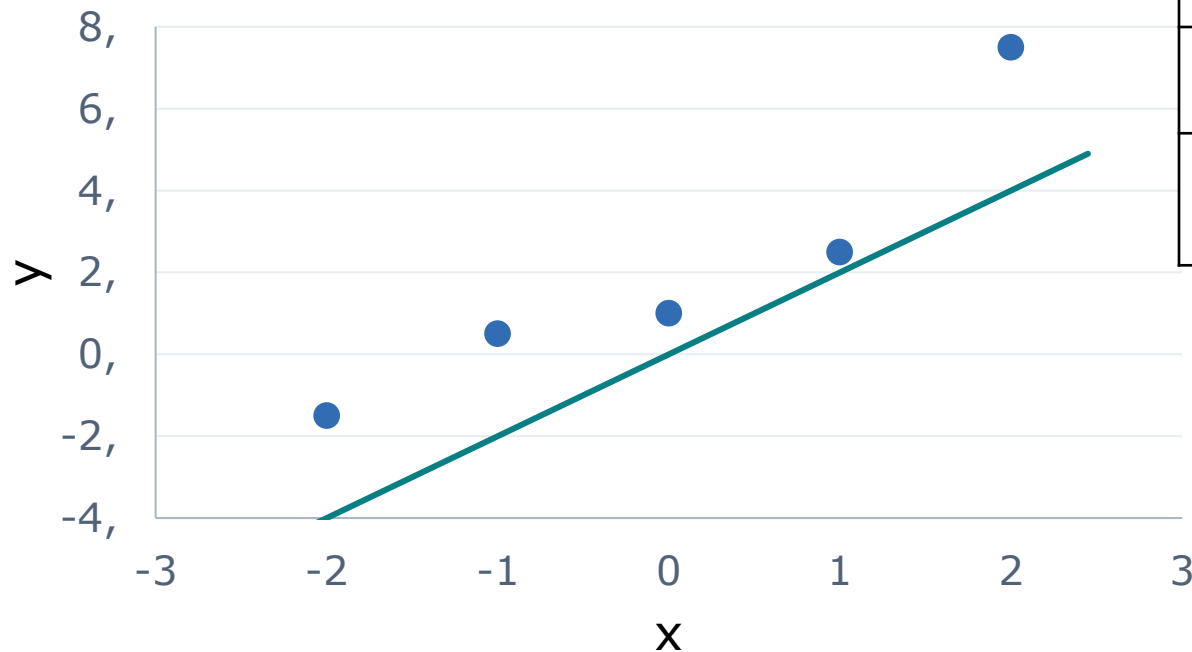
$$\left(\mathcal{L}_{\text{sq}}^{\text{slope}}\right)'(a) = 0 \quad \Leftrightarrow \quad a \sum_{i=1}^N x_i^2 = \sum_{i=1}^N x_i y_i$$

$$a^* = \frac{\sum_{i=1}^N x_i y_i}{\sum_{i=1}^N x_i^2} \quad \text{if } \sum_{i=1}^N x_i^2 > 0$$

Again, the loss is an upward-opening parabola.
The unique stationary point is its global minimiser.

If every $x_i = 0$, all slopes give the same predictions and loss.

The fitted line through the origin



Obs	1	2	3	4	5	$\mathcal{L}_{\text{sq}}^{\text{slope}}$
Input	-2	-1	0	1	2	
Target	-1.5	0.5	1	2.5	7.5	
Squared residual	6.25	6.25	1	0.25	12.25	26

$$a^* = \frac{\sum_{i=1}^5 x_i y_i}{\sum_{i=1}^5 x_i^2} = \frac{20}{10} = 2$$

$$\mathcal{L}_{\text{sq}}^{\text{slope}}(a^*) = 26$$

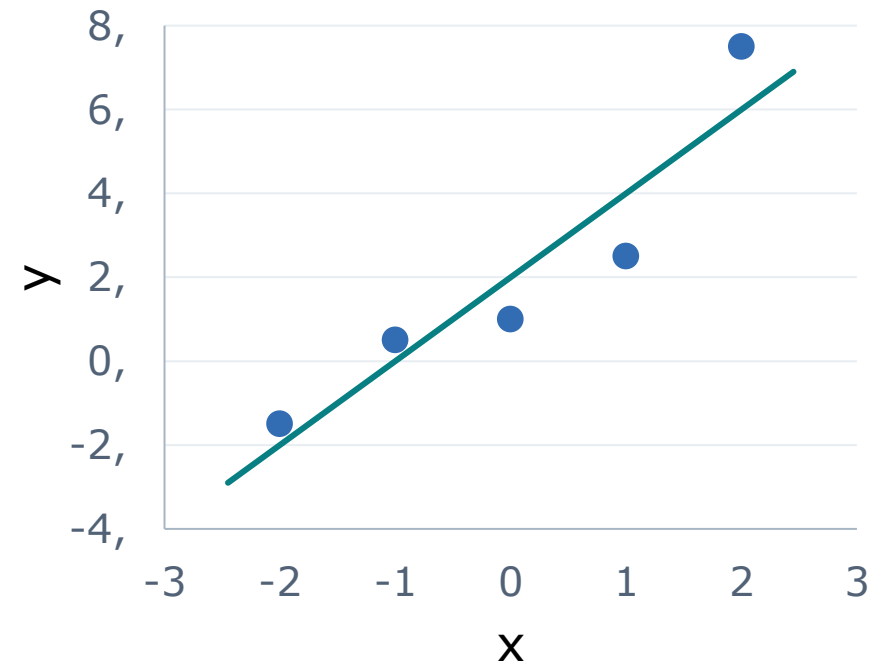
On these data, this fit improves on the best constant, for which $\mathcal{L}_{\text{sq}}^{\text{const}}(b^*) = 46$.

Fitting both slope and intercept

Now both coefficients vary. We collect them in a column vector.

$$\boldsymbol{\theta} = \begin{pmatrix} b \\ a \end{pmatrix}, \quad f_{\boldsymbol{\theta}}(x) = b + ax$$

$$\begin{aligned} \mathcal{L}_{\text{sq}}^{\text{aff}}(b, a) &= \sum_{i=1}^N \ell_{\text{sq}}(f_{\boldsymbol{\theta}}(x_i), y_i) \\ &= \sum_{i=1}^N (b + ax_i - y_i)^2 \end{aligned}$$



The objective is now a function of two parameters.

Affine loss: the intercept derivative

$$\mathcal{L}_{\text{sq}}^{\text{aff}}(b, a) = \sum_{i=1}^N (b + ax_i - y_i)^2 = \sum_{i=1}^N r_i (b, a)^2$$

where $r_i(b, a) = b + ax_i - y_i$ is the i -th residual, so $\frac{\partial r_i}{\partial b}(b, a) = 1$.

The **chain rule**, with outer function $u(r) = r^2$ and inner functions r_i , gives

$$\begin{aligned} \frac{\partial \mathcal{L}_{\text{sq}}^{\text{aff}}}{\partial b}(b, a) &= \sum_{i=1}^N \frac{\partial}{\partial b} (r_i(b, a)^2) = \sum_{i=1}^N 2r_i(b, a) \frac{\partial r_i}{\partial b}(b, a) \\ &= 2 \sum_{i=1}^N (b + ax_i - y_i) = 2 \left(Nb + a \sum_{i=1}^N x_i - \sum_{i=1}^N y_i \right) \end{aligned}$$

Hold a fixed: only b varies. The data x_i and y_i are fixed.

Affine loss: the slope derivative

$$\mathcal{L}_{\text{sq}}^{\text{aff}}(b, a) = \sum_{i=1}^N (b + ax_i - y_i)^2 = \sum_{i=1}^N r_i (b, a)^2$$

where $r_i(b, a) = b + ax_i - y_i$ is the i -th residual, so $\frac{\partial r_i}{\partial a}(b, a) = x_i$.

The **chain rule**, with outer function $u(r) = r^2$ and inner functions r_i , gives

$$\begin{aligned} \frac{\partial \mathcal{L}_{\text{sq}}^{\text{aff}}}{\partial a}(b, a) &= \sum_{i=1}^N \frac{\partial}{\partial a} (r_i(b, a)^2) = \sum_{i=1}^N 2r_i(b, a) \frac{\partial r_i}{\partial a}(b, a) \\ &= 2 \sum_{i=1}^N x_i (b + ax_i - y_i) = 2 \left(b \sum_{i=1}^N x_i + a \sum_{i=1}^N x_i^2 - \sum_{i=1}^N x_i y_i \right) \end{aligned}$$

Hold b fixed: only a varies. The data x_i and y_i are fixed.

The stationarity equations

Set both partial derivatives of $\mathcal{L}_{\text{sq}}^{\text{aff}}$ to zero and divide by two.

$$\frac{\partial \mathcal{L}_{\text{sq}}^{\text{aff}}}{\partial b}(b, a) = 0 \iff Nb + a \sum_{i=1}^N x_i = \sum_{i=1}^N y_i$$

$$\frac{\partial \mathcal{L}_{\text{sq}}^{\text{aff}}}{\partial a}(b, a) = 0 \iff b \sum_{i=1}^N x_i + a \sum_{i=1}^N x_i^2 = \sum_{i=1}^N x_i y_i$$

This is a linear system in the unknowns b and a .

Solving the system

Define the empirical means

$$\bar{x} = \frac{1}{N} \sum_{i=1}^N x_i, \quad \bar{y} = \frac{1}{N} \sum_{i=1}^N y_i$$

The stationarity equations

Set both partial derivatives of $\mathcal{L}_{\text{sq}}^{\text{aff}}$ to zero and divide by two.

$$\frac{\partial \mathcal{L}_{\text{sq}}^{\text{aff}}}{\partial b}(b, a) = 0 \quad \Leftrightarrow \quad N\textcolor{red}{b} + \textcolor{red}{a}N\bar{x} = N\bar{y}$$

$$\frac{\partial \mathcal{L}_{\text{sq}}^{\text{aff}}}{\partial a}(b, a) = 0 \quad \Leftrightarrow \quad \textcolor{red}{b}N\bar{x} + \textcolor{red}{a}\sum_{i=1}^N x_i^2 = \sum_{i=1}^N x_i y_i$$

This is a **linear system** in the **unknowns b and a** .

Solving the system

Define the empirical means

$$\bar{x} = \frac{1}{N} \sum_{i=1}^N x_i, \quad \bar{y} = \frac{1}{N} \sum_{i=1}^N y_i$$

The first equation gives the intercept in terms of the slope.

$$b = \bar{y} - a\bar{x}$$

Substitute this expression into the second equation.

$$a \left(\sum_{i=1}^N x_i^2 - N\bar{x}^2 \right) = \sum_{i=1}^N x_i y_i - N\bar{x} \bar{y}$$

The best slope and intercept

Assume the inputs are not all equal. Then the denominator below is positive.

$$a^* = \frac{\sum_{i=1}^N x_i y_i - N\bar{x}\bar{y}}{\sum_{i=1}^N x_i^2 - N\bar{x}^2} = \frac{\sum_{i=1}^N (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^N (x_i - \bar{x})^2}$$

$$b^* = \bar{y} - a^*\bar{x}$$

We have found a **unique stationary point**.

Is it a global minimum?

Residuals at the stationary point

Let (b^*, a^*) be the stationary point. Its residuals are

$$r_i^* = b^* + a^* x_i - y_i, \quad i = 1, \dots, N.$$

The zero-gradient (stationary point) conditions say precisely that

$$\sum_{i=1}^N r_i^* = 0, \quad \sum_{i=1}^N x_i r_i^* = 0.$$

Any other parameter pair is $(b^* + u, a^* + v)$, with $u, v \in \mathbb{R}$.

The corresponding residuals are

$$(b^* + u) + (a^* + v)x_i - y_i = r_i^* + u + vx_i.$$

A global minimum

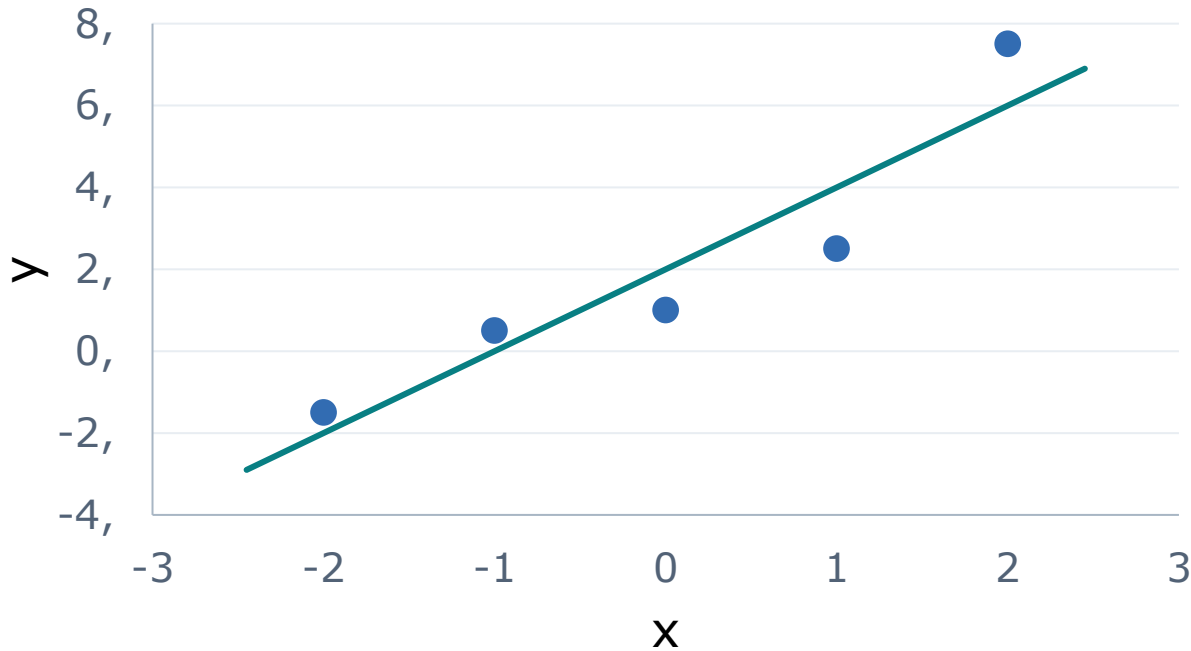
Expanding the squared residuals gives

$$\begin{aligned}\mathcal{L}_{\text{sq}}^{\text{aff}}(b^* + u, a^* + v) &= \sum_{i=1}^N (r_i^* + u + vx_i)^2 \\ &= \mathcal{L}_{\text{sq}}^{\text{aff}}(b^*, a^*) + 2u \underbrace{\sum_{i=1}^N r_i^*}_{=0} + 2v \underbrace{\sum_{i=1}^N x_i r_i^*}_{=0} + \underbrace{\sum_{i=1}^N (u + vx_i)^2}_{\geq 0} \\ &\geq \mathcal{L}_{\text{sq}}^{\text{aff}}(b^*, a^*)\end{aligned}$$

Thus (b^*, a^*) is a global minimiser.

If the inputs are not all equal, equality forces $u = v = 0$: the minimiser is unique.

The affine fit on our data



$$\theta^* = \begin{pmatrix} b^* \\ a^* \end{pmatrix}, \quad f_{\theta^*}(x) = 2 + 2x$$

Obs	1	2	3	4	5	$\mathcal{L}_{\text{sq}}^{\text{aff}}$
Input	-2	-1	0	1	2	
Target	-1.5	0.5	1	2.5	7.5	
Squared residual	0.25	0.25	1	2.25	2.25	6

$$\bar{x} = 0, \quad \bar{y} = 2$$

$$a^* = \frac{\sum_{i=1}^N (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^N (x_i - \bar{x})^2} = \frac{20}{10} = 2,$$
$$b^* = \bar{y} - a^* \bar{x} = 2$$

$$\mathcal{L}_{\text{sq}}^{\text{aff}}(\theta^*) = \mathcal{L}_{\text{sq}}^{\text{aff}}(b^*, a^*) = 6$$

This fit improves on the best slope-only and constant predictors:

$$\mathcal{L}_{\text{sq}}^{\text{slope}}(a^*) = 26, \quad \mathcal{L}_{\text{sq}}^{\text{const}}(b^*) = 46.$$

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An iterative approach

For the affine model, we can calculate the best slope and intercept explicitly.

More complex models may have no explicit solution.
However, we can still use gradients to improve the parameters.

Can we also approach the minimum by improving an initial guess, step by step?

$$\boldsymbol{\theta}^{(0)} = \begin{pmatrix} b^{(0)} \\ a^{(0)} \end{pmatrix}, \quad \boldsymbol{\theta}^{(1)} = \begin{pmatrix} b^{(1)} \\ a^{(1)} \end{pmatrix}, \quad \boldsymbol{\theta}^{(2)} = \begin{pmatrix} b^{(2)} \\ a^{(2)} \end{pmatrix}, \quad \dots$$

We will use the **gradients** to choose each update.

A local linear approximation

For the squared loss $\mathcal{L}_{\text{sq}}^{\text{aff}}$, differentiability gives a local linear approximation.

$$\mathcal{L}_{\text{sq}}^{\text{aff}}(b + \Delta b, a + \Delta a) \approx \mathcal{L}_{\text{sq}}^{\text{aff}}(b, a) + \left(\nabla \mathcal{L}_{\text{sq}}^{\text{aff}}(b, a) \right)^{\top} \begin{pmatrix} \Delta b \\ \Delta a \end{pmatrix}$$

For a sufficiently small step size $\eta > 0$, if we choose

$$\begin{pmatrix} \Delta b \\ \Delta a \end{pmatrix} = -\eta \left(\frac{\partial \mathcal{L}_{\text{sq}}^{\text{aff}}}{\partial b}(b, a), \frac{\partial \mathcal{L}_{\text{sq}}^{\text{aff}}}{\partial a}(b, a) \right)^{\top} = -\eta \nabla \mathcal{L}_{\text{sq}}^{\text{aff}}(b, a)$$

we obtain

$$\mathcal{L}_{\text{sq}}^{\text{aff}}(b + \Delta b, a + \Delta a) \approx \mathcal{L}_{\text{sq}}^{\text{aff}}(b, a) - \eta \left\| \nabla \mathcal{L}_{\text{sq}}^{\text{aff}}(b, a) \right\|_2^2 < \mathcal{L}_{\text{sq}}^{\text{aff}}(b, a)$$

If the gradient is **nonzero**,
a small enough step in its opposite direction **decreases the loss**.

The two updates form a gradient step

The partial derivatives are the components of the gradient.

$$\boldsymbol{\theta} = \begin{pmatrix} b \\ a \end{pmatrix}, \quad \nabla \mathcal{L}_{\text{sq}}^{\text{aff}}(\boldsymbol{\theta}) = \left(\frac{\partial \mathcal{L}_{\text{sq}}^{\text{aff}}}{\partial b}(b, a), \frac{\partial \mathcal{L}_{\text{sq}}^{\text{aff}}}{\partial a}(b, a) \right)^{\top}$$

The two scalar updates give a single displacement in the parameter plane.

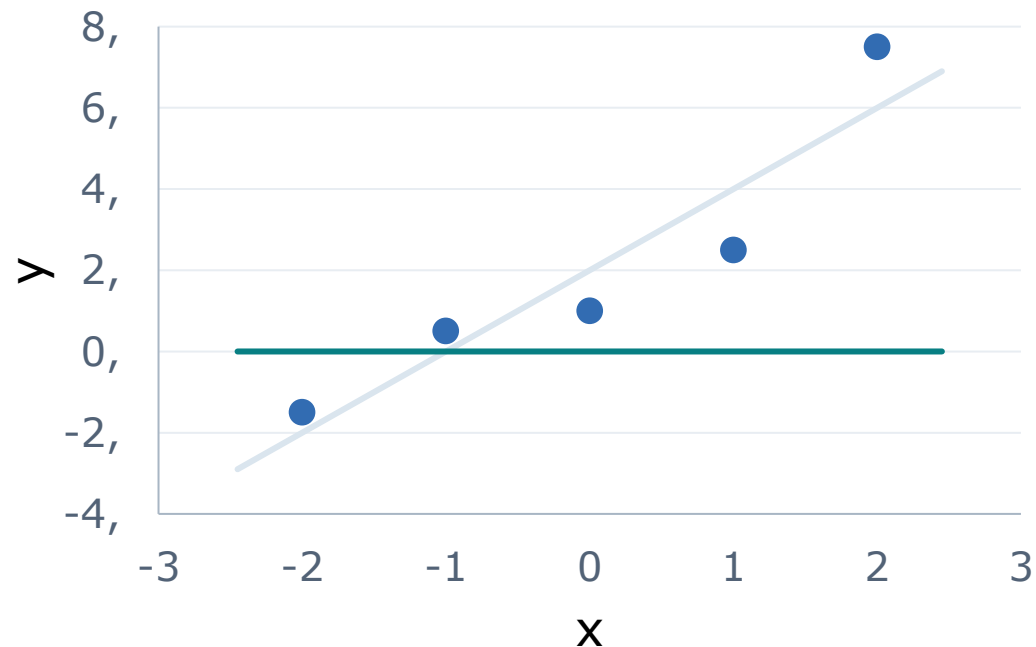
$$\boldsymbol{\theta}^{(t+1)} = \boldsymbol{\theta}^{(t)} - \eta \nabla \mathcal{L}_{\text{sq}}^{\text{aff}}(\boldsymbol{\theta}^{(t)}), \quad \boldsymbol{\theta}^{(t)} = \begin{pmatrix} b^{(t)} \\ a^{(t)} \end{pmatrix}$$

Each step points in the direction of the negative gradient.

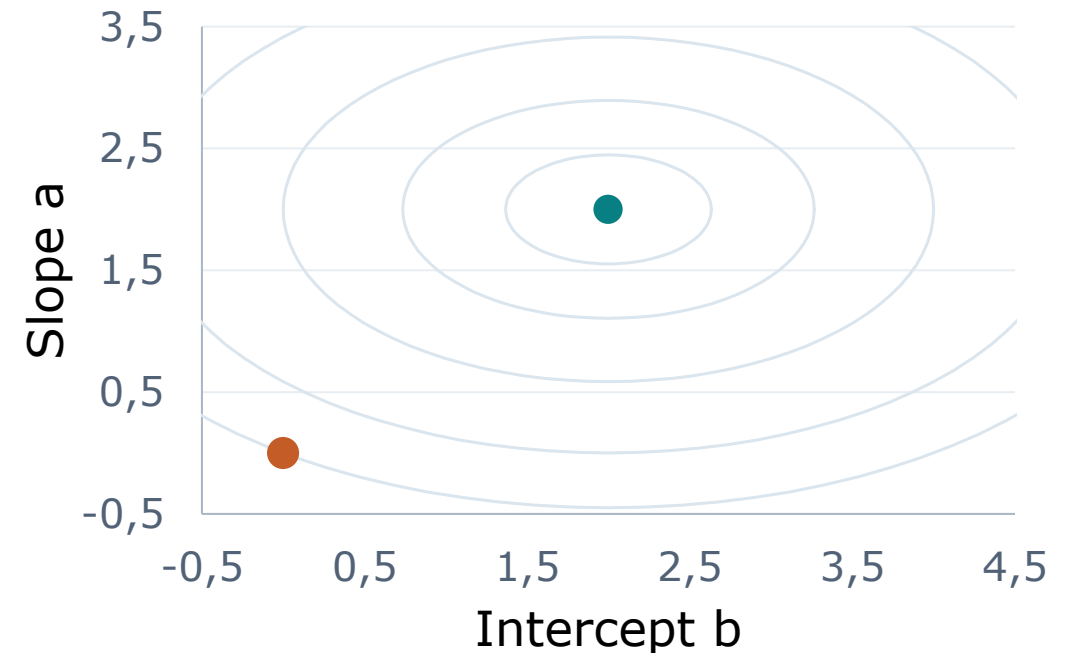
Gradient descent: successive predictors

Same observations, updated coefficients. The learning rate is $\eta = 0.04$.

Predictions on the observed inputs



Path in the parameter plane



$$t = 0 \quad (b^{(0)}, a^{(0)}) = (0,0) \quad \mathcal{L}_{\text{sq}}^{\text{aff}}(\boldsymbol{\theta}^{(0)}) = 66$$

Gradient descent: the first step

Observations	1	2	3	4	5
Input x_i	-2	-1	0	1	2
Target y_i	-1.5	0.5	1	2.5	7.5

$$\boldsymbol{\theta}^{(0)} = (b^{(0)}, a^{(0)})^\top = (0, 0)^\top, \quad \eta = 0.04$$

$$\frac{\partial \mathcal{L}_{\text{sq}}^{\text{aff}}}{\partial b}(0, 0) = -2 \sum_{i=1}^5 y_i = -2 \cdot 10 = -20$$

$$\frac{\partial \mathcal{L}_{\text{sq}}^{\text{aff}}}{\partial a}(0, 0) = -2 \sum_{i=1}^5 x_i y_i = -2 \cdot 20 = -40$$

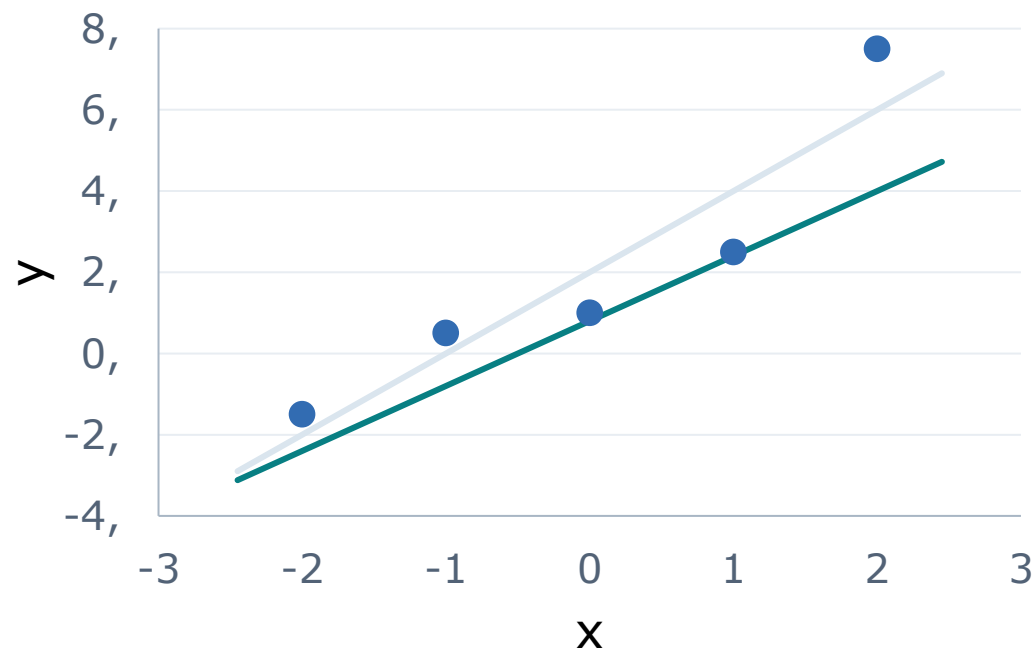
$$\boldsymbol{\theta}^{(1)} = \boldsymbol{\theta}^{(0)} - \eta \nabla \mathcal{L}_{\text{sq}}^{\text{aff}}(\boldsymbol{\theta}^{(0)}) = \begin{pmatrix} 0 \\ 0 \end{pmatrix} - 0.04 \cdot \begin{pmatrix} -20 \\ -40 \end{pmatrix} = \begin{pmatrix} 0.8 \\ 1.6 \end{pmatrix}$$

$$f_{\boldsymbol{\theta}^{(1)}}(x) = 0.8 + 1.6x$$

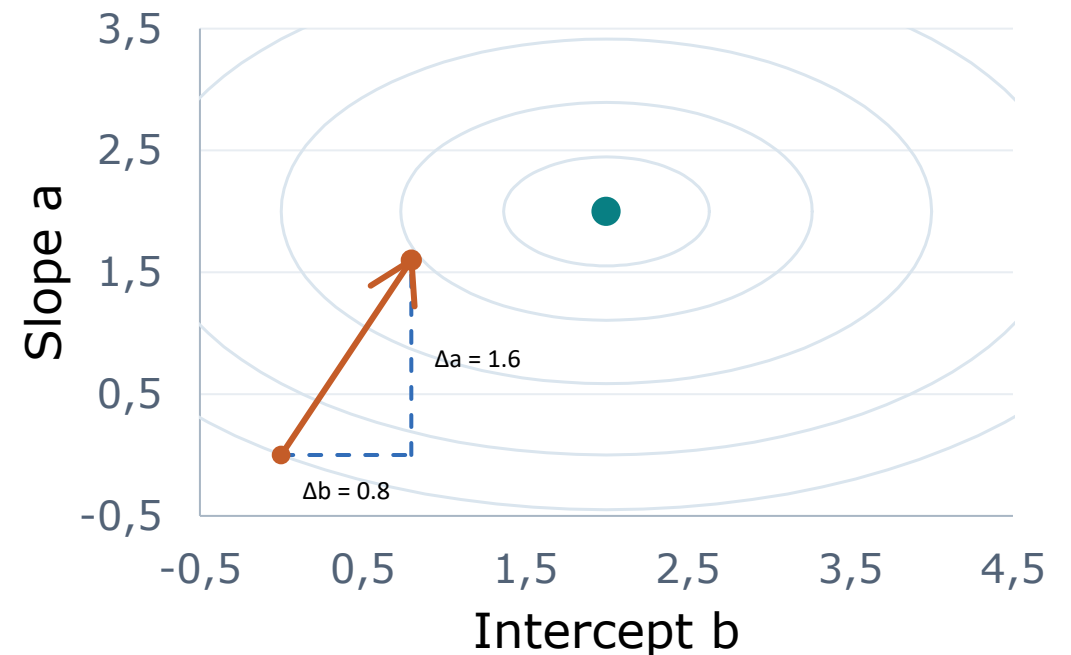
Gradient descent: successive predictors

Same observations, updated coefficients. The learning rate is $\eta = 0.04$.

Predictions on the observed inputs



Path in the parameter plane

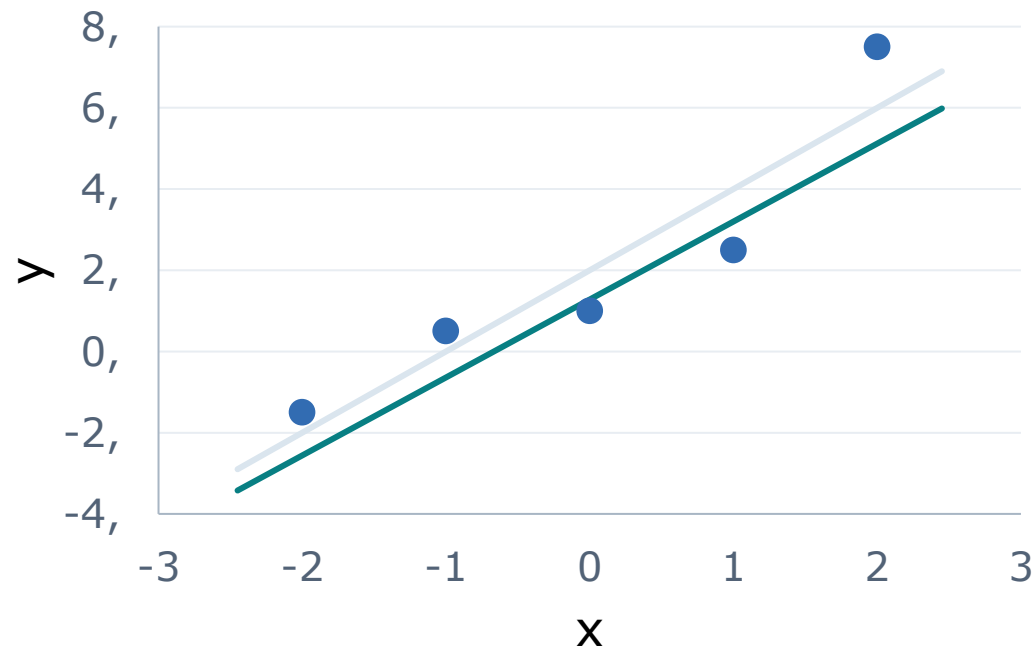


$$t = 1 \quad (b^{(1)}, a^{(1)}) = (0.8, 1.6) \quad \mathcal{L}_{\text{sq}}^{\text{aff}}(\boldsymbol{\theta}^{(1)}) = 14.8$$

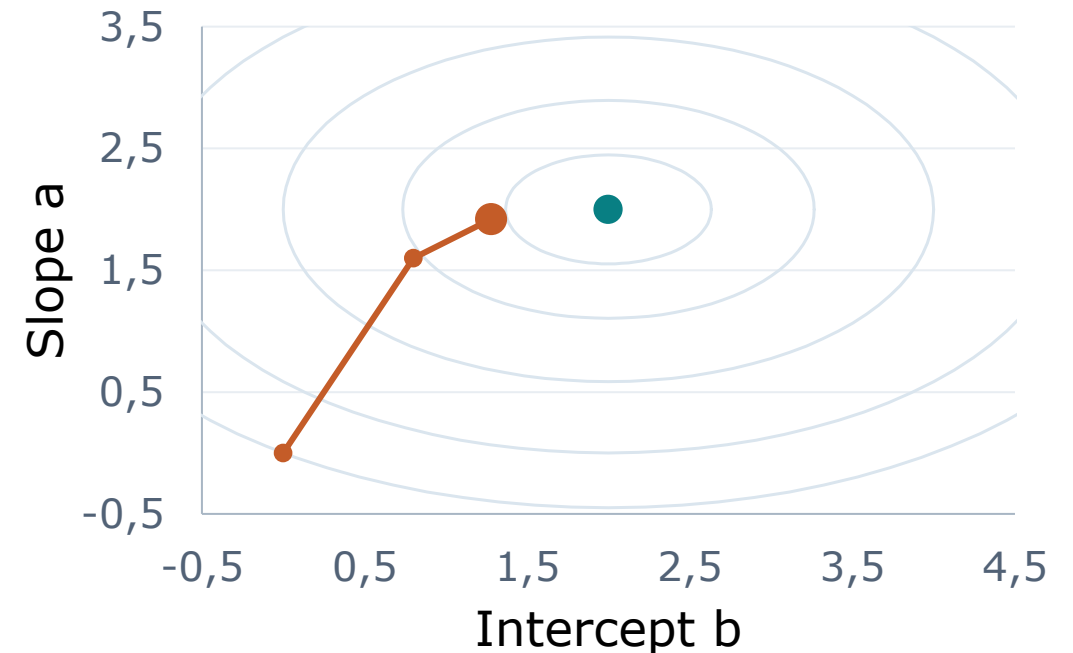
Gradient descent: successive predictors

Same observations, updated coefficients. The learning rate is $\eta = 0.04$.

Predictions on the observed inputs



Path in the parameter plane

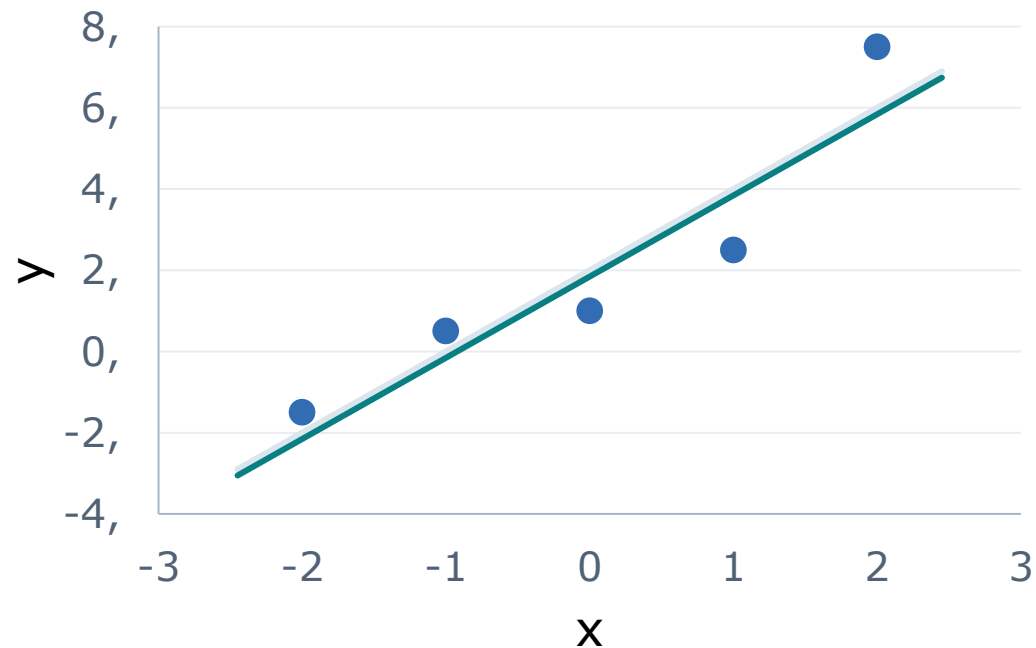


$$t = 2 \quad (b^{(2)}, a^{(2)}) = (1.28, 1.92) \quad \mathcal{L}_{\text{sq}}^{\text{aff}}(\boldsymbol{\theta}^{(2)}) = 8.656$$

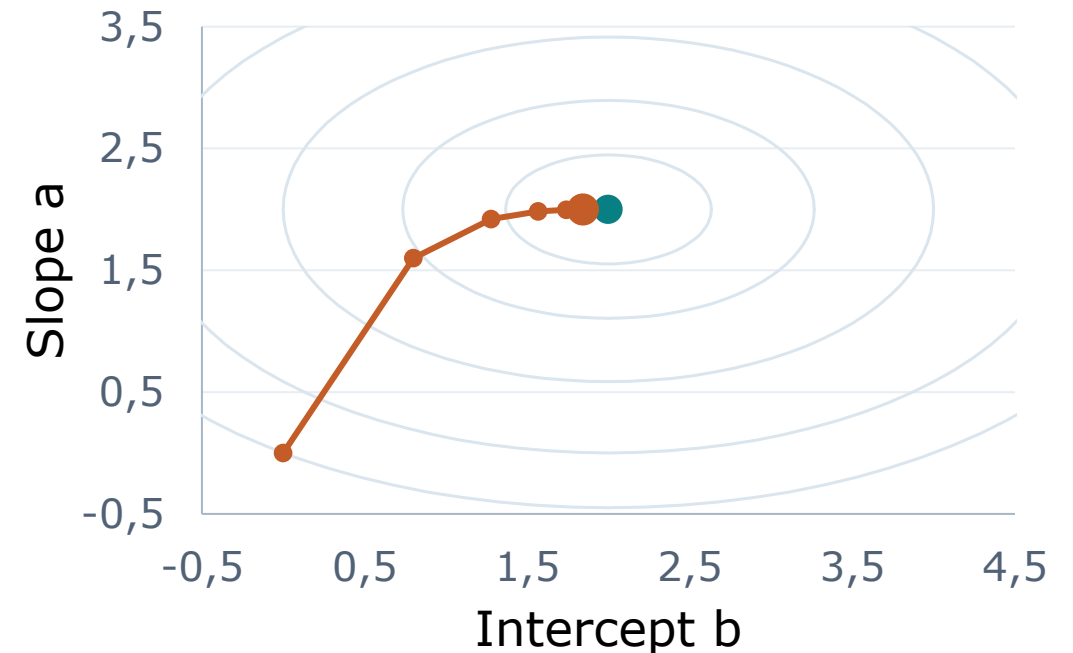
Gradient descent: successive predictors

Same observations, updated coefficients. The learning rate is $\eta = 0.04$.

Predictions on the observed inputs



Path in the parameter plane



$$t = 5 \quad (b^{(5)}, a^{(5)}) = (1.84448, 1.99936) \quad \mathcal{L}_{\text{sq}}^{\text{aff}}(\boldsymbol{\theta}^{(5)}) = 6.12094$$

Today's lecture

Regression and loss functions

Least-squares fitting

Gradient descent

Extensions

Extension 1: nonlinear predictors

So far, our predictors have been affine in the input.
Can we also use predictors that are **nonlinear in the input**?

Example

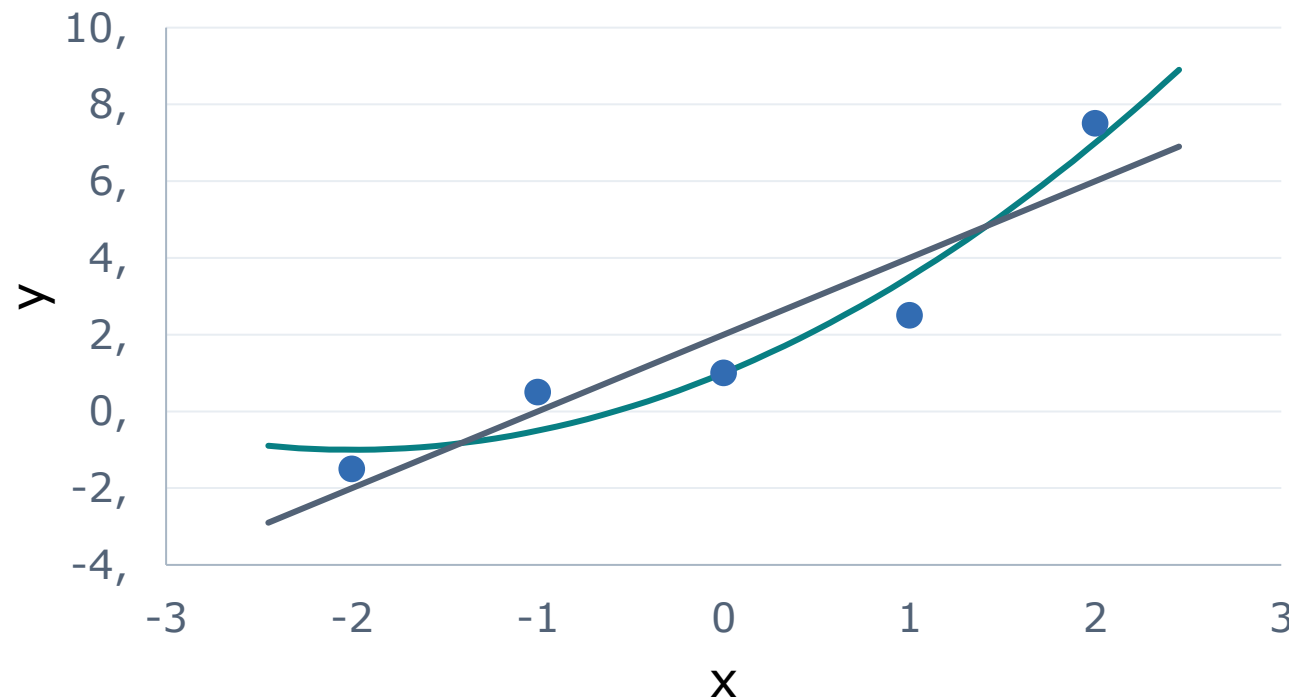
$$f_{\boldsymbol{\theta}}(x) = b + w_1x + w_2x^2, \quad \boldsymbol{\theta} = (b, w_1, w_2)^{\top} \in \mathbb{R}^{3 \times 1}$$

$$\boldsymbol{\Phi}(x) = (1, x, x^2) \in \mathbb{R}^{1 \times 3}, \quad f_{\boldsymbol{\theta}}(x) = \boldsymbol{\Phi}(x)\boldsymbol{\theta}$$

The predictor may be **nonlinear in x**
while remaining **linear in the adjustable coefficients**

A quadratic fit to the same observations

Observations	1	2	3	4	5
Input x_i	-2	-1	0	1	2
Target y_i	-1.5	0.5	1	2.5	7.5



$$\boldsymbol{\phi}(x) = (1, x, x^2)$$

$$\boldsymbol{\theta}^* = (1, 2, 0.5)^\top$$

$$\begin{aligned}\mathcal{L}_{\text{sq}}^{\text{quad}}(\boldsymbol{\theta}^*) &= \sum_{i=1}^N \ell_{\text{sq}}(f_{\boldsymbol{\theta}^*}(x_i), y_i) \\ &= \sum_{i=1}^N (\boldsymbol{\phi}(x_i) \boldsymbol{\theta}^* - y_i)^2 = 2.5\end{aligned}$$

Fixed feature expansion

Choose fixed feature functions $\phi_1, \dots, \phi_m: \mathbb{R} \rightarrow \mathbb{R}$

and define $\boldsymbol{\phi}(x) := (\phi_1(x), \dots, \phi_m(x))$.

For a column vector $\boldsymbol{\theta} = (\theta_1, \dots, \theta_m)^\top \in \mathbb{R}^{m \times 1}$ of parameters, define the predictor

$$f_{\boldsymbol{\theta}}(x) = \boldsymbol{\phi}(x)\boldsymbol{\theta} = \sum_{j=1}^m \theta_j \phi_j(x)$$

Define the feature matrix $\boldsymbol{\Phi} \in \mathbb{R}^{N \times m}$ and the column of observations $\mathbf{y} \in \mathbb{R}^{N \times 1}$ by

$$\Phi_{ij} = \phi_j(x_i), \quad \boldsymbol{\Phi} = \begin{pmatrix} \Phi_{11} & \Phi_{12} & \cdots & \Phi_{1m} \\ \Phi_{21} & \Phi_{22} & \cdots & \Phi_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ \Phi_{N1} & \Phi_{N2} & \cdots & \Phi_{Nm} \end{pmatrix}, \quad \mathbf{y} = \begin{pmatrix} y_1 \\ \vdots \\ y_N \end{pmatrix}$$

Feature expansion: fitting the coefficients

Then the quadratic loss can be expressed as

$$\mathcal{L}_{\text{sq}}(\boldsymbol{\theta}) = \sum_{i=1}^N \ell_{\text{sq}}(f_{\boldsymbol{\theta}}(x_i), y_i) = \|\boldsymbol{\Phi}\boldsymbol{\theta} - \mathbf{y}\|_2^2$$

Gradient

$$\nabla \mathcal{L}_{\text{sq}}(\boldsymbol{\theta}) = 2\boldsymbol{\Phi}^\top (\boldsymbol{\Phi}\boldsymbol{\theta} - \mathbf{y})$$

If the matrix $\boldsymbol{\Phi}^\top \boldsymbol{\Phi}$ is invertible,

The unique stationary point and global minimiser $\boldsymbol{\theta}^*$ is given by

$$\boldsymbol{\theta}^* = (\boldsymbol{\Phi}^\top \boldsymbol{\Phi})^{-1} \boldsymbol{\Phi}^\top \mathbf{y}$$

Gradient descent

$$\boldsymbol{\theta}^{(t+1)} = \boldsymbol{\theta}^{(t)} - \eta \nabla \mathcal{L}_{\text{sq}}(\boldsymbol{\theta}^{(t)}) = \boldsymbol{\theta}^{(t)} - 2\eta \boldsymbol{\Phi}^\top (\boldsymbol{\Phi}\boldsymbol{\theta}^{(t)} - \mathbf{y})$$

Extension 2: several input variables

Can we predict salary y from age, education and years of experience?

An observation may contain several input features.

$$\mathbf{x} = (x_1, \dots, x_d) \in \mathbb{R}^{1 \times d},$$

We associate one weight with each feature, together with a bias (intercept).

$$b \in \mathbb{R}, \quad \mathbf{w} = (w_1, \dots, w_d)^\top \in \mathbb{R}^{d \times 1}, \quad \boldsymbol{\theta} := \begin{pmatrix} b \\ \mathbf{w} \end{pmatrix} \in \mathbb{R}^{(d+1) \times 1}$$

The resulting linear predictor is then

$$f_{\boldsymbol{\theta}}(\mathbf{x}) = (1, \mathbf{x})\boldsymbol{\theta} = b + \mathbf{x}\mathbf{w} = b + \sum_{j=1}^d x_j w_j$$

The design matrix

To handle all N observations $(\mathbf{x}_1, y_1), \dots, (\mathbf{x}_N, y_N)$ at once,
we stack the data into $\mathbf{X} \in \mathbb{R}^{N \times (d+1)}$ and observations into $\mathbf{y} \in \mathbb{R}^{N \times 1}$

$$\mathbf{X} = \begin{pmatrix} 1 & \mathbf{x}_1 \\ \vdots & \vdots \\ 1 & \mathbf{x}_N \end{pmatrix} = \begin{pmatrix} 1 & x_{11} & \cdots & x_{1d} \\ \vdots & \vdots & & \vdots \\ 1 & x_{N1} & \cdots & x_{Nd} \end{pmatrix}, \quad \mathbf{y} = \begin{pmatrix} y_1 \\ \vdots \\ y_N \end{pmatrix}$$

Each row represents one observation.

Each column represents a feature.

The first column contains ones and accounts for the bias.

Several inputs: solution and gradient descent

Then the quadratic loss can be expressed as

$$\mathcal{L}_{\text{sq}}(\boldsymbol{\theta}) = \sum_{n=1}^N \ell_{\text{sq}}(f_{\boldsymbol{\theta}}(\mathbf{x}_n), y_n) = \|\mathbf{X}\boldsymbol{\theta} - \mathbf{y}\|_2^2$$

Gradient and gradient descent

$$\nabla \mathcal{L}_{\text{sq}}(\boldsymbol{\theta}) = 2\mathbf{X}^\top (\mathbf{X}\boldsymbol{\theta} - \mathbf{y})$$

If the matrix $\mathbf{X}^\top \mathbf{X}$ is invertible,

The unique stationary point and global minimiser $\boldsymbol{\theta}^*$ is given by

$$\boldsymbol{\theta}^* = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$$

Gradient descent

$$\boldsymbol{\theta}^{(t+1)} = \boldsymbol{\theta}^{(t)} - \eta \nabla \mathcal{L}_{\text{sq}}(\boldsymbol{\theta}^{(t)}) = \boldsymbol{\theta}^{(t)} - 2\eta \mathbf{X}^\top (\mathbf{X}\boldsymbol{\theta}^{(t)} - \mathbf{y})$$

Lecture notes and exercises

The notes for today's lecture are available on the course GitHub page.

They explain the material in more detail and include exercises.

I will post the exercise solutions before the next lecture.

github.com/semtem0051/2026_27



Scan for course materials